AIAA JOURNAL
Vol. 55, No. 9, September 2017

Modeling Subscale Rotor Wake in Ground Effect with Accurate Turbulent Length Scales

Tarandeep S. Kalra* and James D. Baeder†
University of Maryland, College Park, Maryland 20742

DOI: 10.2514/1.J055647

An accurate prediction of the tip vortex flowfield is required to understand the viscous interactions of the rotor wake and ground prevalent during helicopter brownout. To achieve this goal, a compressible, structured, overset Reynolds-averaged Navier–Stokes-based solver is used to simulate a subscale rotor in ground effect. Novel mesh-gridding techniques are used to avoid prohibitive computational costs. The Reynolds-averaged Navier–Stokes equations solved with the Spalart–Allmaras turbulence model show that a smeared-out flowfield at an early wake ages due to excessive turbulence levels. To further investigate that the source of a smeared rotor wake is the use of turbulence modeling, the computations are performed with laminar flow assumption that show a preserved rotor wake field for longer wake ages. However, an accurate understanding of the cause of brownout requires an accurate prediction of turbulence levels. Therefore, a hybrid Reynolds-averaged Navier–Stokes/large-eddy simulation methodology such as delayed detached-eddy simulation is explored. The flowfield obtained from the delayed detached-eddy simulation methodology still does not preserve the rotor wake. Because the wake capturing grids are highly anisotropic, a modified length scaling is used. The delayed detached-eddy simulation-based length scale with correction for anisotropic grids helps in preserving the rotor wake and shows a close agreement with experimental results.

Downloaded by Indian Institute of Technology, Bombay on October 15, 2025 | http://arc.aiaa.org | DOI: 10.2514/1.J055647

Nomenclature

A	=	area of the rotor blades; πR^2, m^2
C_P	=	power coefficient in-ground effect; power/$(\rho\pi R^2 V_{\text{tip}}^3)$
C_T	=	thrust coefficient in-ground effect; thrust/$(\rho\pi R^2 V_{\text{tip}}^2)$
c	=	rotor blade chord, m
M_{tip}	=	tip Mach number
R	=	radius of the rotor, m
r	=	radial distance, m
r_c	=	vortex core radius, m
T	=	thrust, N
V_{tip}	=	tip speed, m/s
V_θ	=	swirl velocity, m/s
x, y, z	=	Cartesian coordinates
$\Delta x, \Delta y, \Delta z$	=	grid spacing in each coordinate direction
ζ	=	wake age, deg

I. Introduction

HELICOPTERS operating close to the ground in the presence of loose sediment in desert or dusty environments get engulfed in large dust clouds, which is also known as the "brownout" problem. Under certain operational conditions, such as takeoff and landing, an intense dust cloud can severely obscure the pilot's vision, leading to a loss of situational awareness. In the recent past, military operations in brownout conditions have resulted in several helicopter accidents, forcing helicopter analysts to investigate the physics related to brownout and try to develop long-term mitigation strategies. The interaction of the rotor wake with the ground and the dust particles forms the root cause of the brownout problem, and a detailed understanding of the fluid physics along with a good knowledge of the fluid–particle interaction under these conditions can help researchers develop effective means of preventing and/or mitigating the adverse effects of rotorcraft brownout.

To understand the effect of rotor wake interaction with the ground in the context of the brownout problem, several experimental analyses have been performed. Most of these efforts have been with the use of laboratory/microscale rotors. On several occasions, the experiments have been conducted without the dust particles to have a better understanding of rotor wake interaction with ground. One such experimental study was performed by Lee et al. [1] using particle image velocimetry (PIV) measurements on a microscale rotor at various heights above a ground plane: $0.25R$, $0.5R$, $1.0R$, and $1.5R$, where R is the rotor radius. As the rotor wake approached the ground, vortex stretching was shown to have the effect of increasing swirl velocities within the vortex cores, leading to a reintensification of vorticity. It was found that these stretching effects were largest when the rotor hovered at an intermediate height above the ground. Beyond this height, the effects of vortex diffusion became more significant than vortex stretching.

Milluzzo et al. [2] measured the evolution of a hovering subscale rotor wake in-ground effect (IGE) for various tip shapes (rectangular, swept, British Experimental Rotor Programme-like, and slotted-tip) using flow visualization and phase-averaged PIV techniques. The rotor operated at a $1R$ rotor height above the ground. The measurements resulted in a detailed dataset of rotor wake characteristics such as core size and swirl velocity profiles for an up to 800 deg wake age.

Nathan and Green [3] studied both single- and dual-phase flowfields around a microscale rotor IGE operating at different heights above ground and at low advance ratios. Investigation using single-phase flow showed the formation of a recirculation region at low advance ratios and the formation of a ground vortex at higher advance ratios. Additional experiments conducted by Johnson et al. [4] and Sydney et al. [5] on a hovering microrotor showed the phenomenon of entrainment and upliftment of sediment particles. More recently, Glucksman-Glaser and Jones [6] used dye flow visualization and PIV techniques to qualitatively and quantitatively measure the flowfield around a microscale rotor operating over a sediment bed in a water tank. It was found that the passage of tip vortices was causing the unsteady velocities to cause sediment mobilization.

These aforementioned experiments provide a detailed understanding of the fluid dynamics for the micro- and subscale rotor in-ground effect. However, the wide range of parameters affecting the brownout problem are still difficult to model in experiments. The main experimental difficulty comes in simulating the brownout conditions in

Received 31 August 2016; revision received 4 March 2017; accepted for publication 28 April 2017; published online 6 July 2017. Copyright © 2017 by the American Institute of Aeronautics and Astronautics, Inc. All rights reserved. All requests for copying and permission to reprint should be submitted to CCC at www.copyright.com; employ the ISSN 0001-1452 (print) or 1533-385X (online) to initiate your request. See also AIAA Rights and Permissions www.aiaa.org/randp.

*Scientific Programmer, Integrated Statistics; tarankalra@gmail.com.
†Associate Professor, Department of Aerospace Engineering; baeder@umd.edu.

Downloaded by Indian Institute of Technology, Bombay on October 15, 2025 | http://arc.aiaa.org | DOI: 10.2514/1.J055647

full-scale rotors. Computational studies can be used to overcome these challenges by simulating a wide range of brownout conditions with relative ease. However, one of the challenges in computational studies is to preserve the tip vortex emanating from the blade for a significantly long time to capture their interaction with the ground. These challenges are further aggravated in terms of computational expense when the scale of the problem is increased because finer meshes are required to capture the rotor wake at higher Reynolds number. The focus of several past studies, including this one, is to concentrate on the problem of tracking the evolution of the rotor tip vortices and capture the rotor wake for longer wake ages.

Over the last few years, researchers modeling the rotor wake in the presence of ground have employed aerodynamic models of various levels of sophistication. Most approaches track the tip vortex in the particle-fixed Lagrangian frame of reference [7–9]. Even though these free-vortex-type methods can track the vortex for a long distance, these methods involve a certain degree of empiricism in determining the vortex core radius and rollup. Methods using the vorticity transport model (VTM) [10] or vorticity confinement [11] provide a capability for a more fundamental representation of the formation and evolution of the tip vortex. However, VTM applications have been run as an inviscid model. On the other hand, vorticity confinement applications have used very coarse meshes that may not be sufficient to capture the details of the vortical wake. Furthermore, in these studies, the blades were treated using lifting-line aerodynamics with their accompanying assumptions. Thus, in their current forms, they still retain a fair amount of empiricism.

To avoid the empiricism present in the aforementioned models, several researchers have employed a Reynolds-averaged Navier–Stokes (RANS) methodology to capture the rotor wake in the presence of ground. Hariharan et al. [12] used a RANS-based solver, called Helios, that was capable of adaptive mesh refinement to resolve the important flow features locally along with a combination of higher-order schemes. This work used an inviscid wall boundary condition to model the ground plane; hence, the turbulence near the ground was not accurately captured.

Kutz el al. [13] performed rotor IGE computations in hover and low-speed forward flight. Computations were performed using a RANS solver with the ground plane using a viscous wall boundary condition. The lack of discrete vortices was observed in the flowfield due to the lack of mesh resolution in this work.

In the aforementioned efforts to use a RANS methodology, the use of fine meshes or viscous boundary conditions has been avoided to prevent computational expense. To reduce the computational expense of a RANS methodology, Thomas et al. [14] developed a hybrid methodology that used a free vortex methods (FVM) (Lagrangian model) to obtain the rotor wake and convect the vortex filaments, coupled with a RANS (Eulerian model) model close to the ground to capture the turbulence near the ground. The results showed promise in avoiding huge computational costs of resolving the rotor wake all the way to the ground; however, the use of FVM methods still leads to empiricism.

All the aforementioned modeling efforts either comprised the use of empiricism to model rotor wake or did not have sufficient mesh resolution to resolve the rotor wake accurately. The current work seeks to overcome these issues using a RANS-based computational fluid dynamics (CFD) solver. The mesh resolution is provided with the use of parallelization and intelligent clustering of mesh points in regions of interest. The current work builds upon a framework described by Lakshminarayan et al. [15] to model a hovering microrotor (aspect ratio of 4.387) in ground effect with high fidelity. This work modeled the experimental work of Lee et al. [1] in IGE conditions at different rotor heights above ground and provided a highly accurate comparison of the performance and rotor wake between the experiments and simulations. The current work extends the computational framework presented in [15] to model a subscale rotor (aspect ratio of 9.132) operating at a rotor height of $1R$ above ground. The experiments of Milluzzo et al. [2] as described in the next section are used for validation. A novel mesh setup is formed for these high-aspect-ratio rotor blades to provide a balance between the available computational resources without sacrificing the accuracy of capturing a rotor wake for longer wake ages. New challenges with the RANS methodology are observed and hybrid RANS/large-eddy simulation (LES) methods are explored that provide good agreement with the available subscale rotor experiments [2]. The following section describes the experimental setup for the subscale rotor blade.

II. Experimental Setup for Validation

The experimental setup of Milluzzo et al. [2] shown in Fig. 1 is used to validate the flowfield of a hovering one-bladed rotor operating in ground effect. The motivation of the experimental work to use a one-bladed rotor hovering in ground effect stems from focusing on the generation and convection of the rotor wake, for which the properties can be more accurately measured until it interacts with the ground with fewer blades. This allows for a better understanding of aperiodicity of the rotor wake at older wake ages. The rotor consisted of an untwisted rectangular blade with a NACA 2415 airfoil. The rotor blade had a chord length of 44.5 mm and a radius of 0.408 m, resulting in an aspect ratio of 9.132. The blades were operated at a collective pitch of 4.5 deg. The rotor operated at a tip Mach number of 0.24 and a chord Reynolds number of 250,000. The rotor blades operated at a rotor height of $1R$ above the ground. The experiments were performed on four different blades obtained by varying the blade tip shapes. In this work, only the rectangular tip blade is modeled to understand the importance of length scale selection in turbulence models for preserving vorticity in sections below.

III. Computational Setup: Flow Solver

In the present work, the RANS-based solver OVERTURNS [16] is used to perform the computations in a time-accurate manner in an inertial frame of reference. The code solves the compressible RANS equations using a preconditioned dual-time scheme. The second-order backward differencing scheme is used for time marching, and the linear system is solved using the lower–upper symmetric Gauss–Seidel [17] framework. Low Mach preconditioning [18] is used to improve the convergence and accuracy of the flow solver. A third-order MUSCL scheme [19] using Koren's limiter [20] is used for spatial reconstruction on the blade and the main background mesh, whereas a fifth-order mapped, weighted essentially nonoscillatory scheme [21] scheme is used for the vortex-tracking grids and the funnel-shaped mesh. More details about the meshes are provided in the following section. The Roe flux difference scheme [22] is used for calculating inviscid fluxes, and a second-order central scheme is used to discretize the viscous terms. The Spalart–Allmaras [23] turbulence model is employed for the RANS closure. The production term in this eddy viscosity model is modified to account for the reduction in

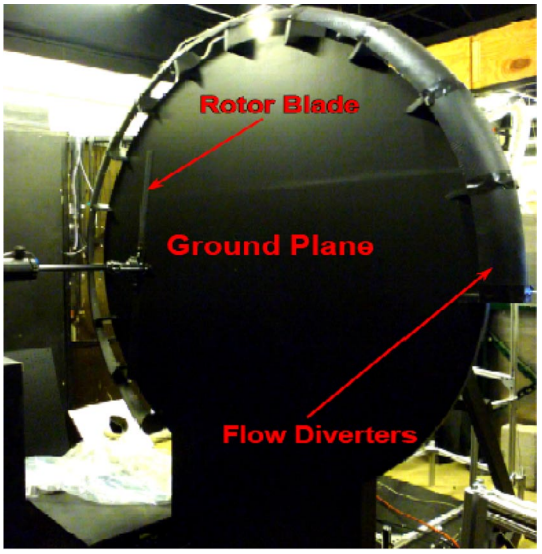
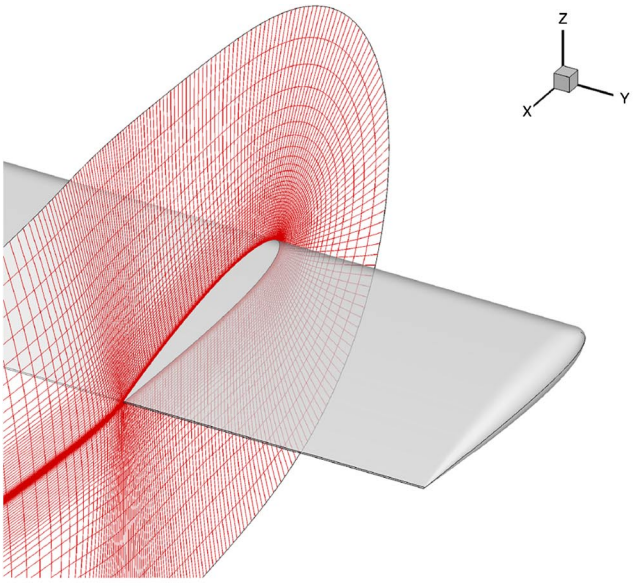


Fig. 1 Experimental setup of one-bladed subscale rotor of Milluzzo et al. [2].

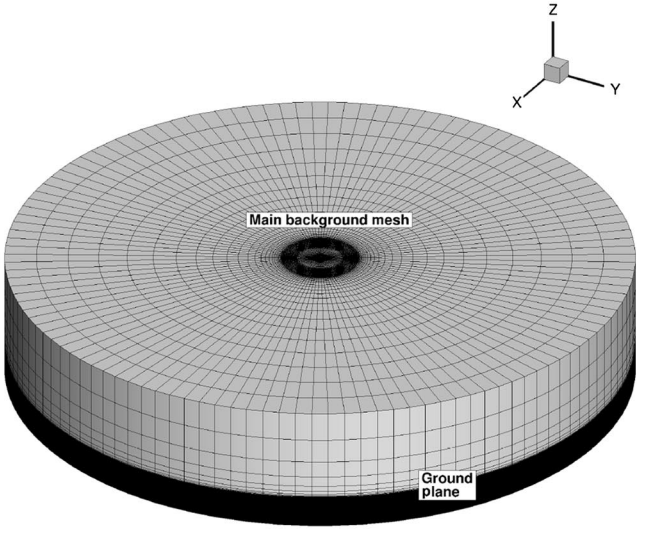
turbulence levels at the vortex core due to rotational effects [24]. The implicit hole-cutting technique, developed by Lee [25] and improved by Lakshminarayan [16], is used to determine the connectivity information between the various overset meshes.

IV. Computational Setup: Mesh System

The wake capturing is initially performed by using a blade mesh and a single cylindrical background mesh. The blade mesh comprises a body-conforming C-O type mesh (Fig. 2a). The blade mesh consists of $267 \times 185 \times 50$ points in the streamwise, spanwise, and wall-normal directions. One-hundred and eighty-seven mesh points are along the airfoil surface in the streamwise direction. A spacing of 5.0×10^5 chords is used in the wall-normal direction, which corresponds to a $y+$ value of around 1.0. The background mesh (Fig. 2b) is a cylindrical Cartesian mesh that consists of identical planes rotated in the azimuthal direction. Each azimuthal plane is spaced at every 1.5 deg in azimuth. A cross-sectional view of the background mesh is shown in (Fig. 3a). In these most refined regions, the background mesh has a grid spacing of 0.04 chords in both the radial and vertical directions. The mesh extends to $10.0R$ in the radial direction. The total number of mesh points in the background cylindrical mesh is $247 \times 207 \times 205$. The simulation is performed for 15 revolutions. After 15 revolutions, the flowfield is well developed.



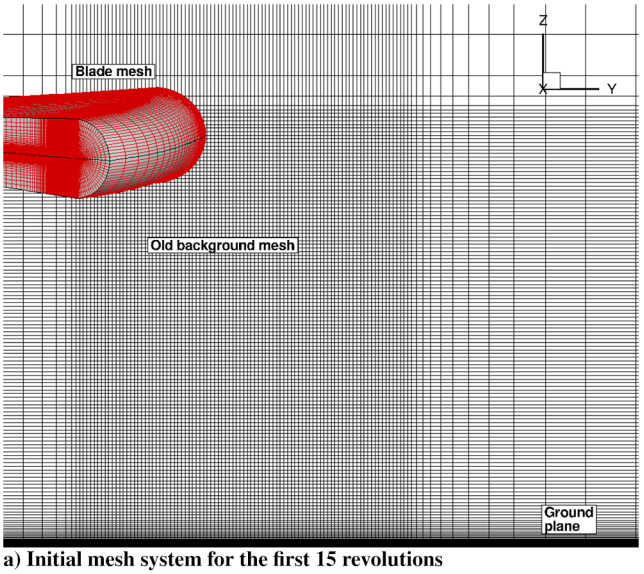
a) Body-conforming blade mesh



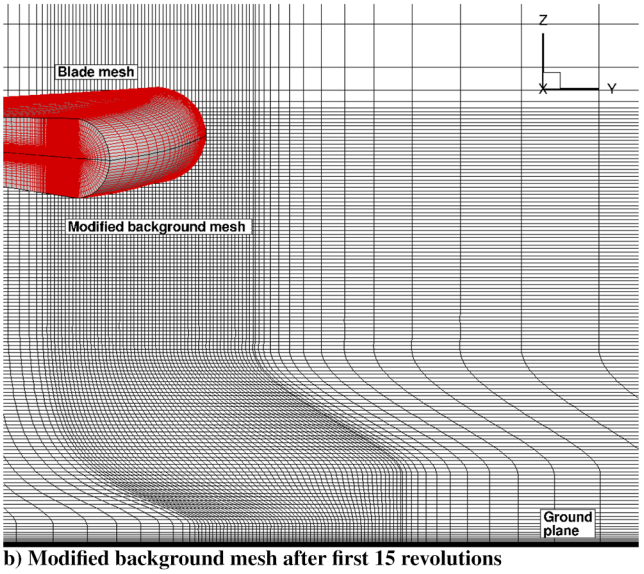
b) Main background mesh with overset meshes

Fig. 2 Mesh system without vortex tracking grid.

It is then used to obtain the instantaneous tip vortex trajectory by searching for the maximum vorticity magnitude corresponding to the vortex center at all wake ages. Once the tip vortex trajectory is obtained, the background mesh is modified to follow the path of the rotor wake to reduce the number of mesh points in the radial direction. The resulting modified background mesh is shown in Fig. 3b and is referred to as the main background mesh. In addition to the strategic refinement of the main background mesh, a vortex tracking grid (VTG), seen in Fig. 4a, is used to better the preserve the tip vortices along the path of the rotor wake. The VTG is a three-dimensional helical mesh, which is generated by azimuthally extruding a two-dimensional (2-D) Cartesian plane along the instantaneous tip vortex trajectory. These 2-D planes are placed at every 1.0 deg of the azimuth, which is two-thirds the azimuthal distance in the background mesh. The 2-D planar section of the VTG is a Cartesian mesh with equispaced mesh points near the middle, which stretches at the edges to reduce the cell size disparity between the VTG and the main background mesh. The extent of the 2-D plane is calculated by accounting for the expected wandering of the tip vortex. At the edge of the VTG, the cell spacing is equal to that in the new background mesh; whereas at the center, the cell spacing is approximately one-fourth. If the VTG is obtained by using an instantaneous tip vortex trajectory, the resulting grid contains irregularities along the azimuth direction. These irregularities are damped out by using a Laplacian-based smoothing.



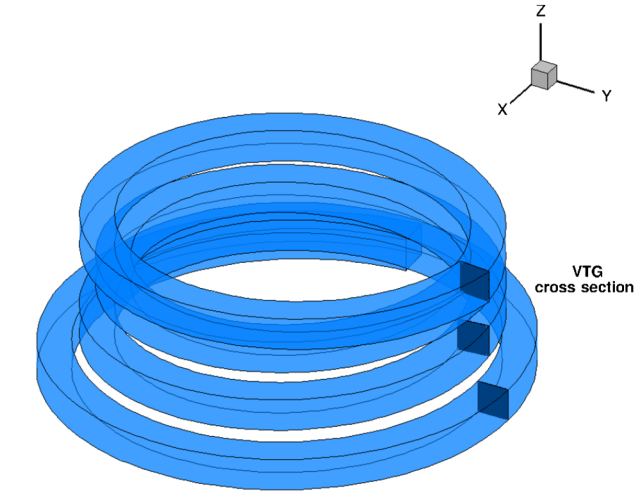
a) Initial mesh system for the first 15 revolutions



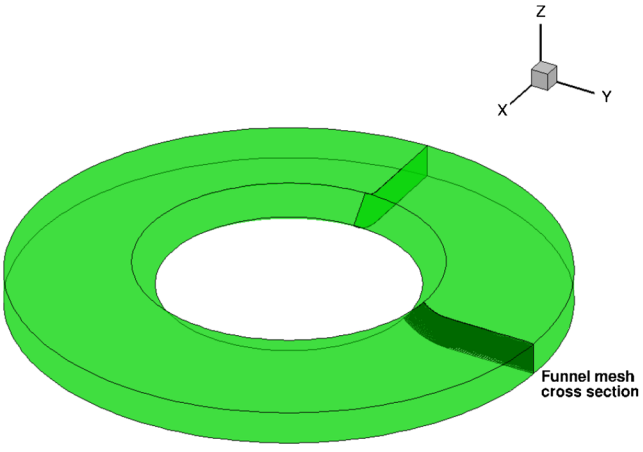
b) Modified background mesh after first 15 revolutions

Fig. 3 Main background mesh modification.

Downloaded by Indian Institute of Technology, Bombay on October 15, 2025 | http://arc.aiaa.org | DOI: 10.2514/1.J055647



a) Vortex tracking grid



b) Funnel mesh

Fig. 4 Additional overset meshes.

The resolution of the vortices close to the ground is further improved by using a funnel mesh as in Fig. 4b. The cross-sectional limits of the funnel mesh are based on the instantaneous tip vortex trajectory. It can be observed that the mesh is following a radially outward path as the vortices move radially outward close to the ground. The cell spacing in the azimuth direction is two-thirds of the main background mesh. The radial and vertical directions are refined twice as much (i.e., $0.2c$) as the main background mesh.

The combination of the blade mesh, main background mesh, vortex tracking grids, and funnel mesh shown in Fig. 5 form the complete mesh system for wake capturing. Table 1 shows the number of points in each type of mesh for the one-bladed subscale rotor cases.

V. Computational Setup Parameters and Costs

The chosen time step corresponds to 0.25 of azimuth for all the cases. A constant Courant–Friedrichs–Lewy number is maintained along with a pseudotime-step size of 10 for the Newton subiterations to provide stability throughout the iterations. In addition, at each time step, six subiterations are used to remove linearization errors. The preconditioning Mach number is set to the local Mach number, which is contrary to the tip Mach number used for the microscale rotor simulations as described by Lakshminarayan et al. [15]. This is seen to improve convergence for the subscale rotor cases. Each simulation is run for a total of 19 revolutions. The first 15 revolutions only use the modified main background mesh and the funnel mesh, and then the vortex tracking grids are added for another four revolutions. The VTGs are modified at each revolution during these last four revolutions. The calculations take about 6.25 days to complete 19 revolutions (i.e., 3.04 revolutions per day) when run in parallel on

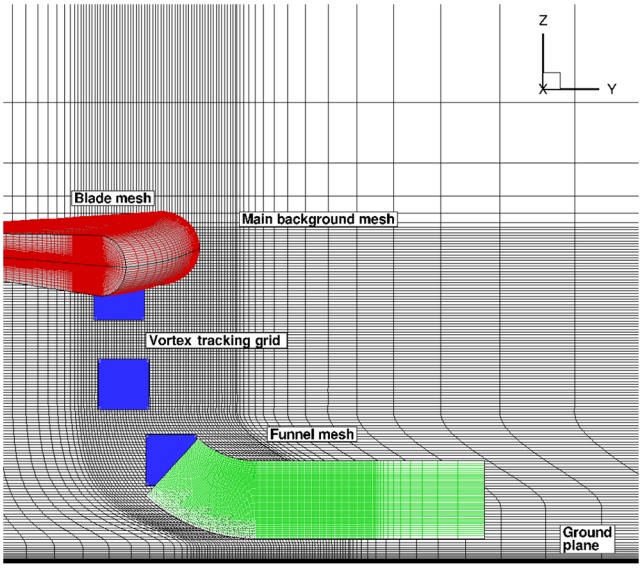


Fig. 5 Complete mesh system.

32 AMD Interlagos Opteron 2.3 GHz processors. A total of 240 processors were used in each of these simulations.

VI. Results

Four different cases are discussed in the following section in order to understand the importance of appropriate turbulent length scaling for the subscale rotor simulations. The baseline case involves the use of RANS methodology previously used by Lakshminarayan et al. [15] successfully for the microscale rotor.

VII. Case 1: Baseline Case with S–A Model

The baseline case consists of the RANS-based solver with the Spalart–Allmaras (S–A) model to model the subscale rotor. The experimental value of thrust coefficient C_T is 0.0018. The computational prediction with the baseline S–A turbulence model predicts a C_T of 0.0019. The computed power coefficient C_P is 0.00012. The experiments do not predict a C_P value.

Figure 6a shows the comparison of the experimental measurements and computational results for the variation of the peak swirl velocity with wake age. The peak swirl velocity is nondimensionalized by the tip speed of the rotor blade. The experiments show a decrease in the peak swirl velocity until a wake age of 360 deg. This is followed by a slight increase in the peak swirl velocity due to the in-ground effect. Compared to the experiments, the computations show a rapid decrease in peak swirl velocity with increasing wake age. Even at an early wake age of 180 deg, the peak swirl velocity predicted by the computations is twice as low as the experimental values.

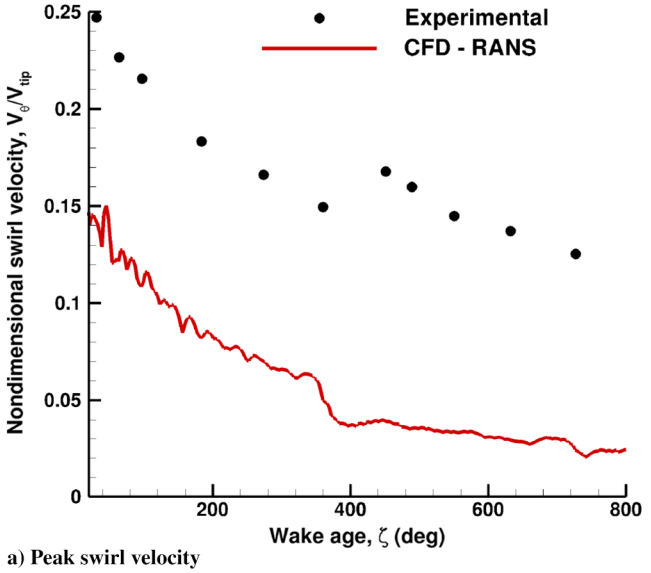
Figure 6b shows the variation of the vortex core radius as a function of wake age for the different tip shapes. The core radius is nondimensionalized by the rotor blade chord. It can be noticed that the core radius grows much faster in the computations as compared to the experimental results. For instance, the core radius at the 720 wake age in the computations grows to an order of 0.3 chords, whereas the experiment only has a core size on the order 0.075 chords.

The results from the peak swirl velocity and core radius values show that the tip vortices are not being captured well, even at early

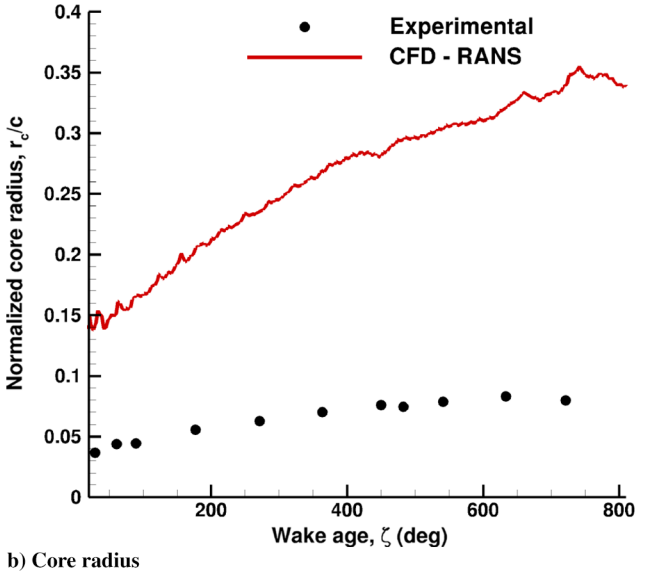
Table 1 Number of mesh points for subscale rotor

Mesh	Dimensions	Mesh points (in millions)
Blade mesh	$267 \times 185 \times 51$	2.52
Main background mesh	$247 \times 241 \times 341$	19.11
Vortex tracking grid	$145 \times 145 \times 1336$	28.21
Funnel mesh	$367 \times 100 \times 325$	11.96
Total	— —	61.82

Downloaded by Indian Institute of Technology, Bombay on October 15, 2025 | http://arc.aiaa.org | DOI: 10.2514/1.J055647

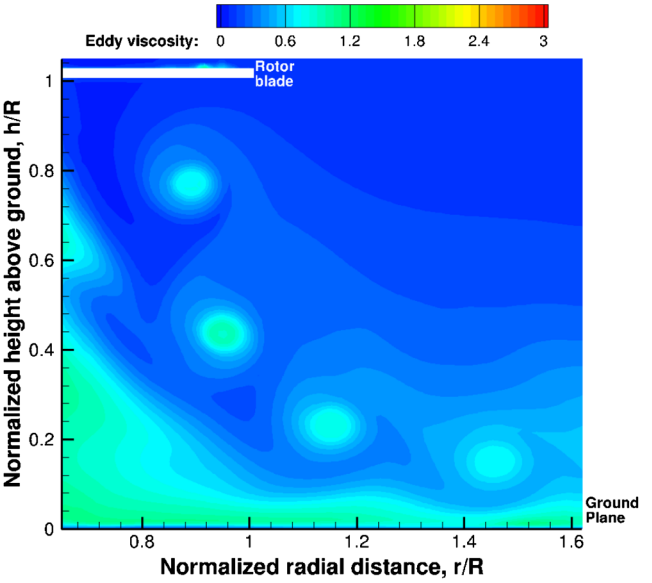


a) Peak swirl velocity

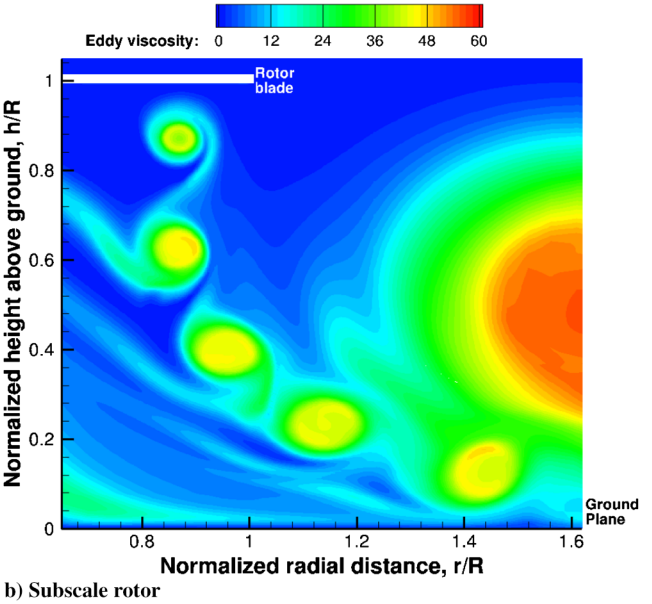


b) Core radius

Fig. 6 Comparison of CFD (RANS) predicted tip vortex characteristics: a) peak swirl velocity comparison and b) core radius comparison.



a) Microscale rotor



b) Subscale rotor

Fig. 7 Comparison of CFD (RANS) predicted eddy viscosity contours at an azimuthal plane of 0 deg.

wake ages, in the RANS computations; thus, further investigation of the CFD methodology is necessary. In the earlier work by Lakshminarayan et al. [15], the microscale rotor regime simulations were validated with an excellent agreement with available experimental results. However, the subscale rotor predictions do not show a good agreement with the experimental results, and the vortices are seen to diffuse too much at early wake ages. Attempts made to refine the mesh resolution by twice the size in each direction of the vortex tracking grids (results not shown) do not improve in preserving the rotor wake strength, even at early wake ages. This indicated that the use of the turbulence model in its current form may be a source of excessive smearing of the rotor wake. This issue is examined by comparing the eddy viscosity contours for the subscale regime as compared to the microscale rotor regime (operating at $1R$ rotor height). Figure 7 shows the computed eddy viscosity contours at a wake age of 0 deg for both the microscale rotor and the subscale rotor, respectively. Note that the subscale simulations show the presence of a starting vortex. It can be observed that the flow is close to laminar for the microscale rotor, except for the region close to the ground. However, the subscale rotor computations show a much more highly turbulent flowfield. One can even observe high levels of eddy viscosity at the center of the vortex cores; whereas theoretically, the vortex cores are expected to be laminar in nature [26]. This implies that the turbulence model (i.e., the Spalart–Allmaras turbulence model with rotational correction) being used in the RANS equations to solve for the eddy viscosity values is leading to an excessive diffusion of the vortex core. It is a known fact that the RANS equations with the Spalart–Allmaras turbulence model were formulated to predict attached or thin shear-layer flows well. This causes modeling limitations when predicting flows that involve large amounts of flow separation.

As seen above, the CFD predicted rotor wake for the sub scale rotor is diffused at early wake ages. To identify that the source of dissipation of vortices is the use of turbulence modeling, simulations are performed with the assumption of laminar flow i.e. (turbulence model is not used). The hypothesis of the laminar flow assumption-based simulations is to avoid the high turbulence levels and show a dependency on the choice of turbulence modeling in preserving the rotor wake without any increase in mesh resolution. The laminar flow simulations are performed on the wake capturing meshes (modified background mesh, vortex tracking grids, and funnel mesh). The blade mesh still solves for turbulent parameters using the S–A turbulence model. The results from the revised computations with laminar flow assumption on wake capturing meshes form case 2, as described in the following.

Downloaded by Indian Institute of Technology, Bombay on October 15, 2025 | http://arc.aiaa.org | DOI: 10.2514/1.J055647

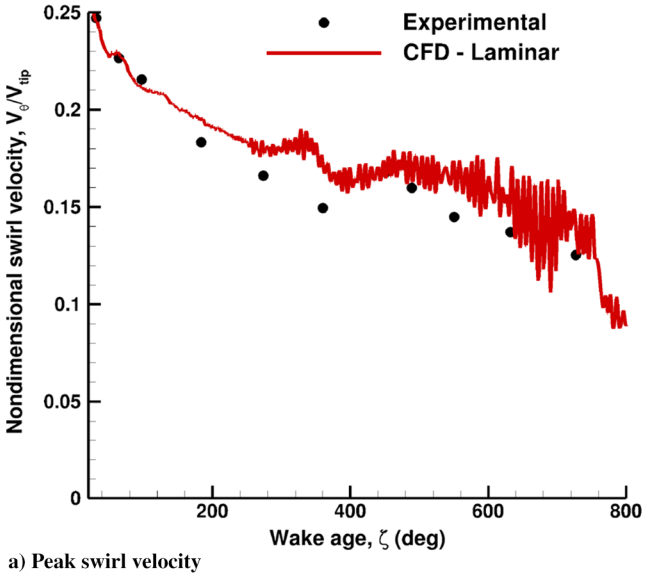
Downloaded by Indian Institute of Technology, Bombay on October 15, 2025 | http://arc.aiaa.org | DOI: 10.2514/1.J055647

VIII. Case 2: Laminar Flow Assumption on Wake-Capturing Meshes

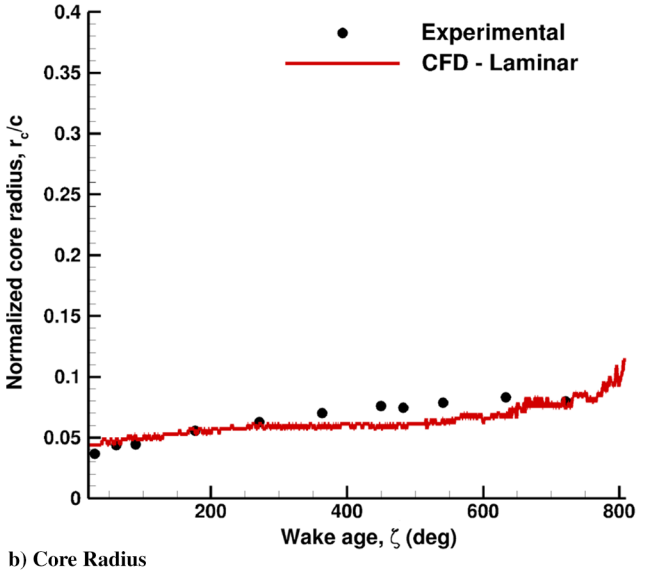
The simulations are performed with the laminar flow assumption on the wake capturing meshes as described previously. The values of C_T and C_P do not change as compared to case 1.

Figure 8a compares the peak swirl velocity from case 2 with a laminar flow assumption with varying wake age. In the previous computations (i.e., case 1) with the turbulence model being used in all the meshes, the peak swirl velocity decreases at a high rate and reaches very low values at older wake ages. However, in case 2 with the laminar flow assumption on the wake capturing meshes, the peak swirl velocity values look in good agreement with the experiments.

Figure 8b shows the core radius growth from case 2 and is compared with the experiments. The computations in case 2 show a close agreement with the experiments. This shows that the assumption of laminar flow in wake capturing meshes is preserving the vortex for longer wake ages. Consequently, these results confirm that the amount of turbulence predicted by the use of S–A model in case 1 were leading to the excessive diffusion of the tip vortices. It can also be noticed that the revised computations with laminar assumption on the wake capturing meshes show a smaller core radius growth than the experiments due to the lack of any physical turbulence. This also indicates that the use of a laminar flow assumption might be overpredicting the vortex strength in case 2 as compared to the experiments. More important, the experimental work (Milluzzo et al. [2], Johnson et al. [4], and Sydney et al. [5]) to study brownout indicates that the entrainment of sediment during brownout is caused by the viscous interaction of the rotor wake and ground. This includes the presence of discrete tip vortices trailed from rotor blades as well as secondary flows from a vortex pairing that create an unsteady aerodynamic environment near the ground, causing the sediment entrainment. This necessitates the accurate prediction of turbulence levels at the ground plane for an accurate understanding of the physical phenomenon affecting the mobilization of the particles during brownout.



a) Peak swirl velocity



b) Core Radius

Fig. 8 Comparison of CFD (laminar flow assumption on wake capturing meshes) predicted tip vortex characteristics: a) peak swirl velocity comparison and b) core radius comparison.

IX. Case 3: Delayed Detached-Eddy Simulations, Use of a RANS/LES Approach

Based on the previous two cases of computations, it is established that the use of a turbulence model in its current form is the cause of excessive diffusion of the tip vortices. Moreover, accurate prediction of turbulence levels is required in the computations in order to understand the interaction of the rotor wake with the ground plane with ramifications to the brownout phenomenon. It is a known fact that the RANS equations were formulated to predict attached or thin shear-layer flows well. This causes modeling limitations using the RANS equations when predicting flows that involve large amounts of flow separation. Therefore, one needs to investigate the use of a higher-fidelity approach such as large-eddy simulation for this problem. Because the LES approach is seen to be prohibitively expensive for practical engineering applications, one way to use a higher level of approximation to account for turbulence above that in RANS is to use a hybrid approach between LES and RANS. In a hybrid approach, the boundary-layer/near-wall regions (small-scale turbulence) are modeled using RANS, and regions of massive separation (large-scale turbulence) are resolved using LES. The existing RANS methodology can be extended to include the hybrid approach by modifying the distance function in the S–A model. The modified distance function is based on grid spacing to provide the switch between the RANS and LES limits. Accordingly, a grid spacing-based limiter approach was developed by Spalart et al. [27] and is referred to as the detached-eddy simulation (DES) [27]. To apply DES, a simple modification to the distance function d is made in the S–A turbulence model that depends upon both the wall distance and the local grid spacing. The modified distance function is as follows:

$$\tilde{d} = \min(d, C_{\text{DES}}\Delta) \tag{1}$$

where $C_{\text{DES}} = 0.65$ and Δ is given by the following:

$$\Delta = \max(\Delta x, \Delta y, \Delta z) \tag{2}$$

and where Δx, Δy, and Δz are the grid spacings in all three directions. Spalart et al. [28] mentioned that the application of DES in cases with thick boundary layers or shallow separation regions caused the local grid spacing, even inside the boundary layer, to get chosen from the modified distance function; this resulted in an early onset of the LES mode. This caused a reduction in the eddy viscosity levels, and this phenomena is termed as "modeled stress depletion." To overcome this issue in DES, Spalart et al. [28] introduced the delayed detached-eddy simulation (DDES). The modified distance function based on DDES depends on the following:

$$\tilde{d} = d - f_d \max(d, C_{\text{DES}}\Delta) \tag{3}$$

where

$$f_d = 1 - \tan h([8r_d]^3)$$
$$r_d = \frac{\nu_t + \nu}{(U_{i,j}U_{i,j})^{0.5}k^2d^2} \tag{4}$$

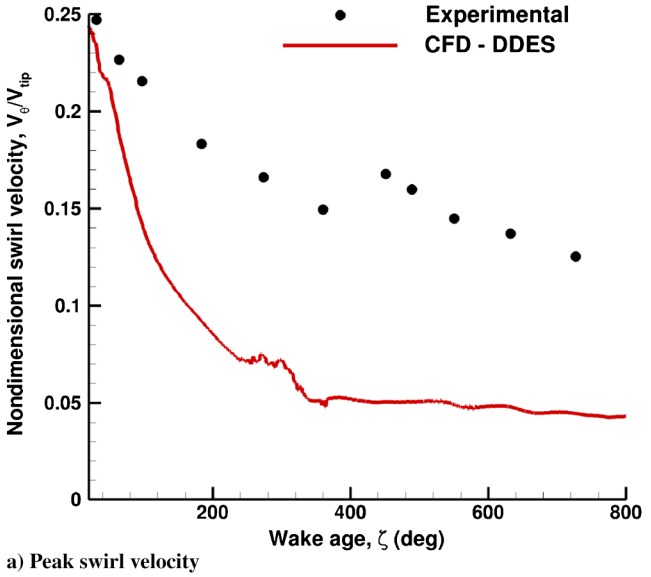
where $U_{i,j}U_{i,j}$ is the inner product of the velocity gradient tensor, k is the Kármán constant, ν_t is the kinematic eddy viscosity, ν is the

molecular eddy viscosity, and d is the normal distance to the wall. Such a formulation results in r_d dropping to zero toward the edge of the boundary layer, giving f_d a value of one there, and thus transitioning to the LES mode. Inside the logarithmic layer, $r_d = 1$, making $f_d = 0$, and ensuring that the RANS mode is activated.

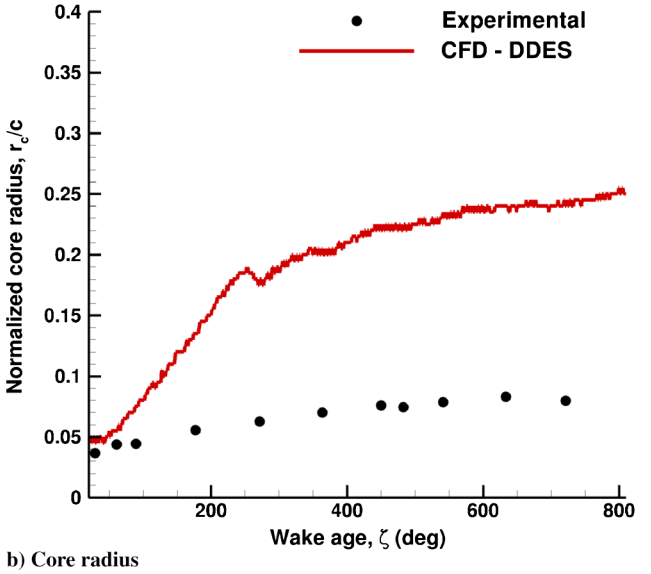
This modified distance function has been implemented into the S–A turbulence model in the OVERTURNS solver, and the computations for the subscale rotor are subsequently revised. The results from the revised computations are referred to as obtained from the S–A/DDES methodology. This modification to the S–A turbulence model is used for the blade mesh along with the wake capturing meshes, and the results are described in the following.

The values of C_T and C_P do not change as compared to case 1. The tip vortex characteristics are first studied by comparing the peak swirl velocity (Fig. 9a) with varying wake ages. The peak swirl velocity with the DDES based length scale modification in computations shows a close agreement with experiments up until 25 deg of wake age but begins to drop sharply after that. Beyond a wake age of 400 deg, the computational values become constant but are three times lower than that of the experiments.

Figure 9b shows the core radius growth from case 3 in comparison with the experiments. Similar to the pattern of peak swirl velocity, the core radius values are observed to grow rapidly beyond a wake age of 25. These results from the peak swirl velocity and core radius show that the DDES based length scale modification to turbulence model for computations show a marginal improvement over RANS computations and do not suffice to preserve the rotor wake. This leads to the discussion in the next section that highlights the importance of the choice of turbulent length scale for hybrid RANS/LES methods such as DDES.



a) Peak swirl velocity

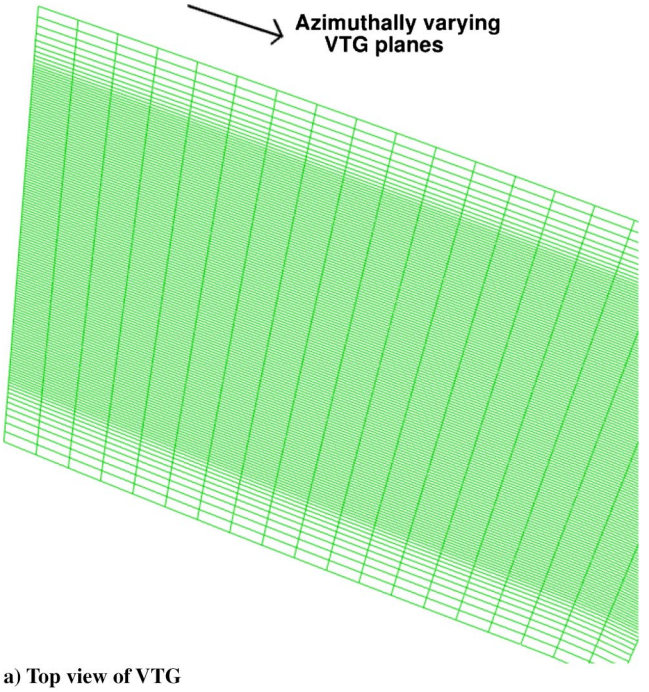


b) Core radius

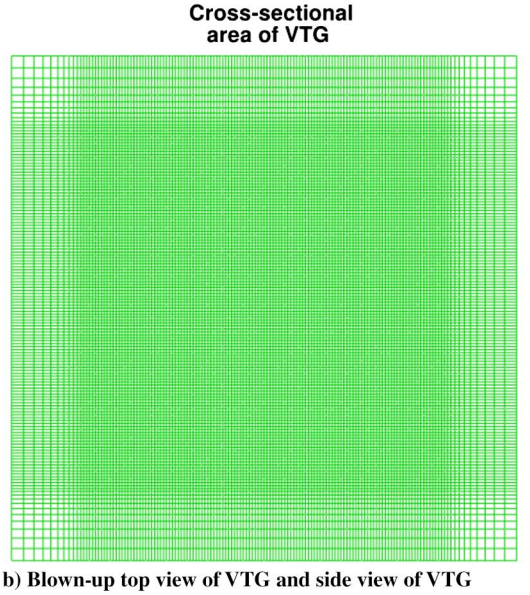
Fig. 9 Comparison of CFD (DDES) predicted tip vortex characteristics: a) peak swirl velocity comparison and b) core radius comparison.

X. Case 4: Delayed Detached-Eddy Simulations, Correcting for Anisotropic Grids

The results presented in the preceding section show that the DDES formulation improves the capturing of tip vortex characteristics marginally. An important factor to consider in the DDES formulation is the length scale function Δ, that is based on the maximum of the grid spacing and is designed to work well for isotropic grids. However, in the current work, the grid spacing is highly anisotropic, especially for the vortex tracking grids. This can be observed from Figs. 10a and 10b,



a) Top view of VTG



b) Blown-up top view of VTG and side view of VTG

Fig. 10 Figures depicting anisotropy of vortex tracking grids.

Downloaded by Indian Institute of Technology, Bombay on October 15, 2025 | http://arc.aiaa.org | DOI: 10.2514/1.J055647

which show the top and side views of the vortex tracking grids. The minimum grid spacing in the VTGs for the subscale rotor consists of a spacing of $0.01c$ in the vertical and radial directions, whereas the maximum grid spacing in the azimuthal direction is approximately 0.16 chords (azimuth of one; as seen in Fig. 10a). This leads to a maximum value of the aspect ratio for the VTGs of 16:1. This implies that the isotropic definition of the Δ function used in the original DDES formulation would not apply for such a highly anisotropic grid system. Scotti et al. [29] defined a modified value of the Δ function that accounts for anisotropy of the grids and is defined as follows:

$$\Delta = f(a_1, a_2) \times \max(\Delta x, \Delta y, \Delta z)^{1/3} \tag{5}$$

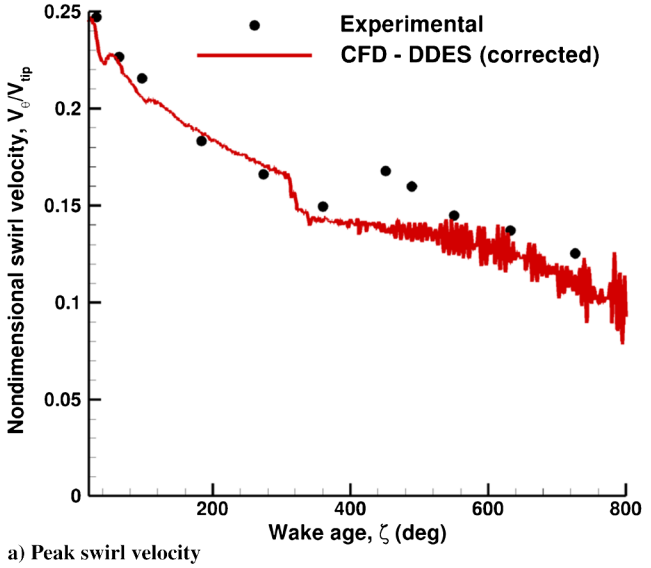
$$f(a_1, a_2) \approx \cosh \sqrt{\frac{4}{27}[(\ln(a_1))^2 - \ln(a_1)\ln(a_2) + \ln(a_2)^2)]} \tag{6}$$

where a_1 and a_2 are the aspect ratios of the grid. The grids have mesh sides (Δx, Δy, Δz) with $\Delta z > \max(\Delta x, \Delta y)$ and aspect ratios of $a_1 = \Delta x/\Delta z$ and $a_2 = \Delta y/\Delta z$.

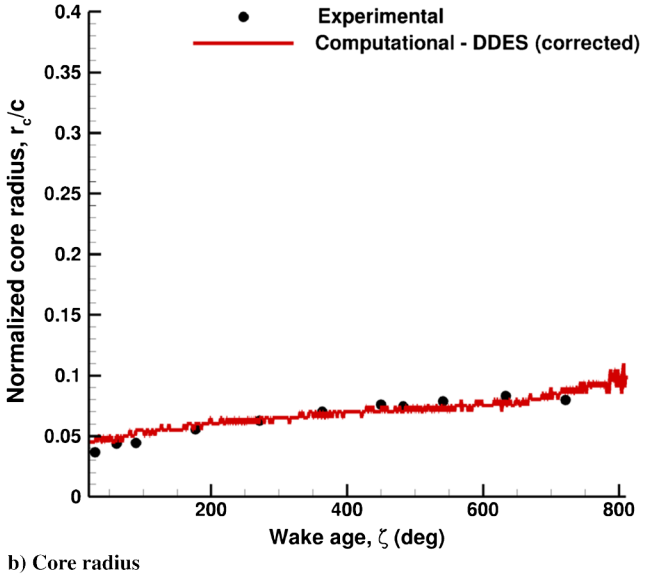
With this implementation, simulations are performed with the anisotropic grid correction implemented in the calculation of the turbulent length scale and specific results from this case (case 4). The values of C_T and C_P do not change as compared to case 1. Figure 11a is first studied by comparing the peak swirl velocity with varying wake ages. In the previous computations (i.e., case 3) with DDES formulation, the peak swirl velocity decreased at a high rate and reached very low values at older wake ages. The peak swirl velocity values in case 4 with the appropriate choice of turbulent length scaling for anisotropic grids shows a good agreement with the experiments. The peak swirl velocity at the wake age of 400 deg increases in the experiments where the vortices are stretched due to ground effect. This increase is not observed in the computational results, although the effect of ground maintains a constant swirl velocity from up until 600 deg, where it starts to decrease again.

The core radius growth from case 4 is compared with the experiments in Fig. 11b. The core radius growth shows a close agreement with the experiments. Note that the underprediction of the core radius growth that was observed in case 2 with the laminar flow assumption on the wake capturing meshes is not observed in case 4.

These results from the DDES formulation with the appropriate length scale show satisfactory results for wake preservation. To further gain confidence of case 4 simulations, one can compare the turbulence

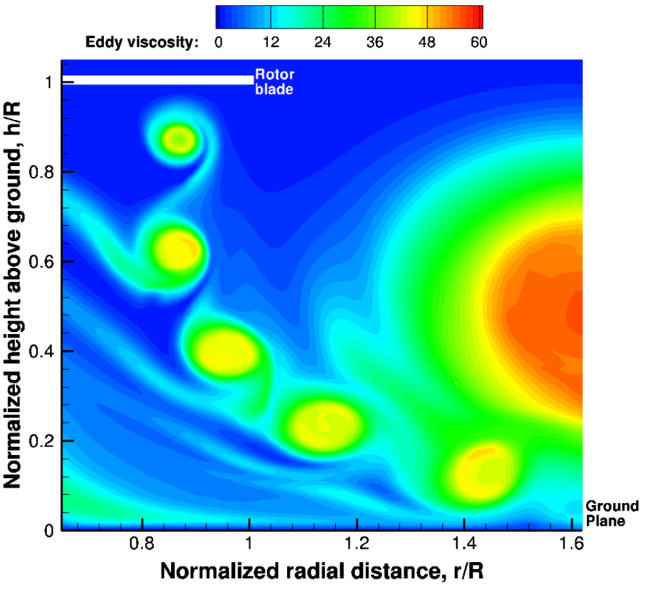


a) Peak swirl velocity

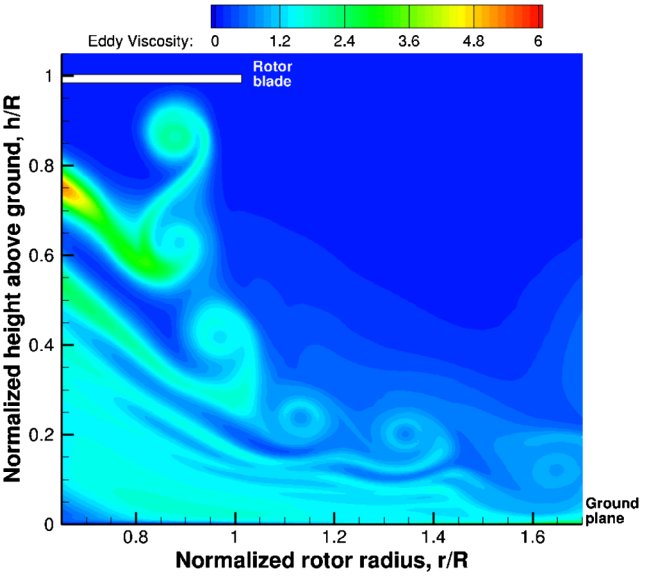


b) Core radius

Fig. 11 Comparison of CFD predicted tip vortex characteristics (DDES with anisotropic grid correction to turbulent length scale): a) peak swirl velocity comparison and b) core radius comparison.



a) Subscale rotor with S–A simulation (case 1)



b) Subscale rotor with SA-DDES simulation (case 4)

Fig. 12 Comparison of CFD predicted eddy viscosity contours at an azimuthal plane of 0 deg for the subscale rotor.

Downloaded by Indian Institute of Technology, Bombay on October 15, 2025 | http://arc.aiaa.org | DOI: 10.2514/1.J055647

levels between the original RANS methodology with the S–A model (case 1) and the S–A/DDES methodology with anisotropic grid correction to the turbulent length scale (case 4). Figure 12 shows the eddy viscosity levels at a wake age of 0 deg for from both the simulations. Note that the levels of eddy viscosity have dropped by almost an order of magnitude. The location of the tip vortices in both the S–A and S–A/DDES results look similar for early wake ages; however, with the additional diffusion occurring in the S–A results, the tip vortex locations start to differ at later wake ages. In addition to that, the vortex centers are seen to be more laminarized, as expected theoretically [26] using the S–A/DDES methodology.

The results of tip vortex characteristics and the analysis of turbulence levels give confidence to modeling the hovering subscale rotor in-ground effect.

XI. Conclusions

The compressible Reynolds-averaged Navier–Stokes solver OVERTURNS was used to simulate a hovering one-bladed subscale rotor operating at a tip Mach number of 0.24 and a tip Reynolds number of 250,000 in-ground effect with the primary objective of capturing the rotor wake accurately for longer wake ages. An earlier framework developed by Lakshminarayan et al. [15] showed a good agreement for the microscale rotor. However, the same framework could not be directly applied to the subscale rotor with the available computational resources. Therefore, the mesh system required novel strategies to efficiently capture the rotor wake. The simulations were done using a system of overset meshes that were strategically placed to accurately capture the formation and evolution of the tip vortex to sufficiently long duration. The results with the RANS solver showed a smeared flowfield and a DES-based approach using an appropriate formulation of the length scale results in accurate flowfield predictions. The following are specific observations/conclusions that can be drawn from the present work:

1) Different cases with different choices of modifications to the turbulence modeling did not change the thrust predictions. This implied that the thrust and power computations were independent from the choice of turbulence modeling.

2) The flowfield generated by RANS simulations showed a highly smeared-out rotor wake at early wake ages. The increased diffusion of tip vortices was attributed to the excessive turbulence levels. This could be observed by looking at the turbulent vortex cores obtained in the RANS simulations.

3) The simulations performed with laminar assumption on wake capturing grids showed a better agreement with the available experiments. However, the lack of turbulence in the flowfield also resulted in artificial preservation of tip vortices, and that led to an overestimation of the vortex strengths in computations as compared with the experiments.

4) The computations were repeated with a hybrid RANS/LES approach referred to as DDES. DDES was seen to marginally improve the CFD predictions from the original RANS methodology.

5) DDES with a modification to the length scale using the correction for anisotropic grids resulted in preserving the flowfield well, and the results agreed well with the experiments. The resulting turbulence levels showed laminar vortex cores, unlike in the RANS simulations.

Acknowledgments

The U.S. Air Force Office of Scientific Research supported this work under a Multidisciplinary University Research Initiative (grant W911NF0410176). The contract monitor was Douglas Smith. The work on the vortex tracking grids was supported by the U.S. Navy's Naval Air Systems Command, with Yik-Loon Lee as the Technical Monitor. The authors wish to thank all of their colleagues at the University of Maryland, the U.S. Government research laboratories, and the rotorcraft industry for the valuable discussions on the brownout problem.

References

[1] Lee, T. E., Leishman, J. G., and Ramasamy, M., "Fluid Dynamics of Interacting Blade Tip Vortices with a Ground Plane," *Journal of the American Helicopter Soc.*, Vol. 55, No. 2, 2010, Paper 022005. doi:10.4050/JAHS.55.022005

[2] Milluzzo, J., "Effect of Blade Tip Shape on Rotor In-Ground-Effect Aerodynamics," Masters Dissertation, Dept. of Aerospace Engineering, Univ. of Maryland, College Park, MD, 2011.

[3] Nathan, N. D., "The Rotor Wake in Ground Effect and its Investigation in a Wind Tunnel," Ph.D. Dissertation, Dept. of Aerospace Engineering, Univ. of Glasgow, Glasgow, 2010.

[4] Johnson, B., Leishman, J. G., and Sydney, A., "Investigation of Sediment Entrainment Using Dual-Phase, High-Speed Particle Image Velocimetry," *Journal of the American Helicopter Society*, Vol. 55, No. 4, 2010, Paper 042003. doi:10.4050/JAHS.55.042003

[5] Sydney, A., "Understanding Brownout using Dual-Phase Particle Image Velocimetry Measurements," *American Helicopter Society 67th Annual Forum Proceedings*, Masters Dissertation, Dept. of Aerospace Engineering, Univ. of Maryland, College Park, MD, 2011.

[6] Glucksman-Glaser, M. S., and Jones, A. R., "Effects of Model Scaling on Sediment Transport in Brownout," *30th AIAA Applied Aerodynamics Conference*, AIAA Paper 2012-2915, June 2012.

[7] Wachspress, D. A., Whitehouse, G. R., Keller, J. D., Yu, K., Gilmore, P., Dorsett, M., and McClure, K., "High Fidelity Rotor Aerodynamic Module for Real Time Rotorcraft Flight Simulation," *American Helicopter Society 65th Annual Forum Proceedings*, Vol. 1, American Helicopter Soc. (AHS), Grapevine, TX, May 2009, pp. 442–462.

[8] Syal, M., and Leishman, J. G., "Modeling of Bombardment Ejections in the Rotorcraft Brownout Problem," *AIAA Journal*, Vol. 51, No. 4, 2013, pp. 849–866. doi:10.2514/1.J051761

[9] Syal, M., and Leishman, J. G., "Predictions of Brownout Dust Clouds Compared to Photogrammetry Measurements," *Journal of the American Helicopter Society*, Vol. 58, No. 2, 2013, pp. 1–18. doi:10.4050/JAHS.58.022001

[10] Phillips, C., and Brown, R. E., "Eulerian Simulation of the Fluid Dynamics of Helicopter Brownout," *Journal of Aircraft*, Vol. 46, No. 4, 2009, pp. 1416–1429. doi:10.2514/1.41999

[11] Wenren, Y., Fan, M., Wang, L., and Steinhoff, J., "Vorticity Confinement and Advanced Rendering to Compute and Visualize Complex Flows," *44th AIAA Aerospace Sciences Meeting and Exhibit*, AIAA Paper 2006-0945, Jan. 2006.

[12] Hariharan, N., Wissink, A., Steffen, M., and Potsdam, M., "Tip Vortex Field Resolution Using an Adaptive Dual-Mesh Computational Paradigm," *49th AIAA Aerospace Sciences Meeting*, AIAA Paper 2011-1108, Jan. 2011.

[13] Kutz, B. M., Kowarsch, U., Kebler, M., and Kramer, E., "Numerical Investigation of Helicopter Rotors in Ground Effect," *30th AIAA Applied Aerodynamics Conference*, AIAA Paper 2012-2913, June 2012.

[14] Thomas, S., Amiraux, M., and Baeder, J. D., "Modeling the Two-Phase Flowfield Beneath a Hovering Rotor on Graphics Processing Units," *AIAA Journal*, Vol. 53, No. 8, 2015, pp. 2300–2320. doi:10.2514/1.J053661

[15] Lakshminarayan, V. K., Kalra, T. S., and Baeder, J. D., "Computational Investigation of a Hovering Microscale Rotor in Ground Effect," *AIAA Journal*, Vol. 51, No. 4, April 2013, pp. 893–909. doi:10.2514/1.J051789

[16] Lakshminarayan, V. K., "Computational Investigation of Micro-Scale Coaxial Rotor Aerodynamics in Hover," Ph.D. Dissertation, Dept. of Aerospace Engineering, Univ. of Maryland, College Park, MD, 2009.

[17] Jameson, A., and Yoon, S., "Lower-Upper Implicit Schemes with Multiple Grids for Euler Equations," *AIAA Journal*, Vol. 25, No. 7, 1987, pp. 929–935. doi:10.2514/3.9724

[18] Turkel, E., "Preconditioning Techniques in Computational Fluid Dynamics," *Annual Review of Fluid Mechanics*, Vol. 31, No. 1, Jan. 1999, pp. 385–416. doi:10.1146/annurev.fluid.31.1.385

[19] Van Leer, B., "Towards the Ultimate Conservative Difference Scheme V. A Second-Order Sequel to Godunovs Method," *Journal of Computational Physics*, Vol. 135, No. 2, 1997, pp. 229–248. doi:10.1006/jcph.1997.5704

[20] Koren, B., "Multigrid and Defect Correction for the Steady Navier–Stokes Equations," *Journal of Computational Physics*, Vol. 87, No. 1, 1990, pp. 25–46.

[21] Henrick, A. K., Aslam, T. D., and Powers, J. M., "Mapped Weighted Essentially Non-Oscillatory Schemes: Achieving Optimal Order Near Critical Points," *Journal of Computational Physics*, Vol. 207, No. 2, 2005, pp. 542–567. doi:10.1016/j.jcp.2005.01.023

Downloaded by Indian Institute of Technology, Bombay on October 15, 2025 | http://arc.aiaa.org | DOI: 10.2514/1.J055647

[22] Roe, P., "Approximate Riemann Solvers, Parameter Vectors and Difference Schemes," *Journal of Computational Physics*, Vol. 135, No. 2, 1997, pp. 250–258.
doi:10.1006/jcph.1997.5705

[23] Spalart, P. R., and Allmaras, S. R., "A One-Equation Turbulence Model for Aerodynamic Flows," *30th AIAA Aerospace Sciences Meeting and Exhibit*, AIAA Paper 1992-0439, Jan. 1992.

[24] Duraisamy, K., "Studies in Tip Vortex Formation, Evolution and Control," Ph.D. Dissertation, Dept. of Aerospace Engineering, Univ. of Maryland, College Park, MD, 2005.

[25] Lee, Y., "On Overset Grids Connectivity and Automated Vortex Tracking in Rotorcraft CFD," Ph.D. Dissertation, Dept. of Aerospace Engineering, Univ. of Maryland, College Park, MD, 2008.

[26] Dacles-Mariani, J., Zilliac, G. G., Chow, J. S., and Bradshaw, P., "Numerical/Experimental Study of a Wingtip Vortex in the Near Field," *AIAA Journal*, Vol. 33, No. 9, 1995, pp. 1561–1568.
doi:10.2514/3.12826

[27] Spalart, P. R., Jou, W.-H., Strelets, M., and Allmaras, S. R., "Comments on the Feasibility of LES for Wings, and on a Hybrid RANS/LES Approach," *Proceedings of the AFOSR International Conference on DNS/LES*, Gayden Press, Ruston, LA, Aug. 1997, pp. 137–147.

[28] Spalart, P. R., Deck, S., Shur, M. L., Squires, K. D., Strelets, M. Kh., and Travin, A., "A New Version of Detached Eddy Simulation, Resistant to Ambiguous Grid Densities," *Theoretical Computational Fluid Dynamics*, Vol. 20, No. 3, 2006, pp. 181–195.
doi:10.1007/s00162-006-0015-0

[29] Scotti, A., Meneveau, C., and Lilly, D. K., "Generalized Smagorinsky Model for Anisotropic Grids," *Physics of Fluid, A*, Vol. 5, No. 9, 1993, pp. 2306–2308.
doi:10.1063/1.858537

S. Fu
Associate Editor

Downloaded by Indian Institute of Technology, Bombay on October 15, 2025 | http://arc.aiaa.org | DOI: 10.2514/1.J055647